mSAC: Enhancing Localization with mmWave Sensing and Orthogonal Signals

Mengning Li
Department of Electrical and Computer Engineering
North Carolina State University
Raleigh, USA
mli55@ncsu.edu

Haocheng Zhu
Department of Electrical and Computer Engineering
The Ohio State University
Columbus, USA
zhu.2991@osu.edu

Wenye Wang
Department of Electrical and Computer Engineering
North Carolina State University
Raleigh, USA
wwang@ncsu.edu

Eylem Ekici
Department of Electrical and Computer Engineering
The Ohio State University
Columbus, USA
ekici.2@osu.edu

IEEE INFOCOM 2025 - IEEE Conference on Computer Communications | 979-8-3315-4305-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/INFOCOM55648.2025.11044619

***Abstract*—mmWave technologies have been adopted into 5G and 6G systems. Current research primarily focuses on enhancing the quality of service and data throughput in mmWave communications, as well as improving the accuracy of sensing from their respective functional perspectives. Consequently, this emphasis often overlooks the potential benefits of integrating communication and sensing, enhancing performance across both functionalities. This paper introduces *mSAC*, a novel mmWave-based system that combines sensing with communication via Orthogonal Frequency-Division Multiplexing (OFDM) signals. This approach differs from traditional mmWave sensing methods that rely on Frequency-Modulated Continuous Wave (FMCW) signals, and it substantially enhances both the accuracy and range of sensing. The *mSAC* system features a Joint Radar-Communication (JRC) transceiver and a communication receiver, incorporating a theoretical model that fuses sensing information from synchronous and asynchronous ends. This model is crucial for smart home applications such as controlling devices, monitoring environmental conditions, and enhancing security. Comprehensive evaluations demonstrate that *mSAC* achieves a 65.93% improvement in sensing accuracy and a 43.19% increase in sensing range compared to conventional methods, thereby setting a new benchmark for integrated sensing and communication systems in mmWave applications.**

***Index Terms*—Millimeter wave radar, Orthogonal frequency-division multiplexing (OFDM), Integrated sensing and communication (ISAC), Wireless sensing.**

I. Introduction

A. Background and Motivation

Integrated sensing and communication (ISAC) unifies both functions into a single system, making it a critical component of modern wireless networks [1]. The recent 802.11bf protocol [2]–[4] exemplifies this by leveraging both low-frequency Wi-Fi and high-frequency millimeter wave bands for sensing. This dual-band approach supports applications in smart home automation and healthcare monitoring, enabling features such as movement detection and vital sign tracking [5]–[9]. The growing adoption of ISAC systems is driven by their ability to maximize resource efficiency and introduce advanced capabilities like environmental awareness and intelligent control, aligning with the demands of next-generation wireless networks.

Current millimeter wave sensing systems primarily utilize frequency-modulated continuous wave (FMCW) signals, achieving high accuracy in various applications [10]. FMCW signals have achieved success in precise localization and behavior sensing [11]. Nevertheless, standard communication protocols predominantly rely on Orthogonal Frequency-Division Multiplexing (OFDM) to ensure high-speed data transmission [12]. OFDM is robust against multipath interference and offers high spectral efficiency, making it essential for modern wireless communication systems. However, leveraging OFDM signals for sensing introduces unique challenges, given that these signals are optimized for communication rather than high-resolution sensing. Overcoming these challenges is critical for realizing the potential of ISAC systems in practical applications.

Since OFDM is primarily a communication signal, it faces several key limitations when adapted for sensing purposes: 1) Unlike FMCW signals, tailored for high-precision measurements, OFDM signals trade off resolution and accuracy when used for sensing. This requires innovative techniques to enhance sensing capabilities without compromising communication efficiency; 2) There is a trade-off between energy consumption and sensing range. Continuously transmitting OFDM signals leads to high energy usage, which is problematic in scenarios where power efficiency is crucial, such as battery-operated devices or large-scale sensor networks [13]. Addressing these limitations is critical for advancing OFDM-based ISAC systems, and this work contributes to this goal by presenting a novel framework that combines sensing and communication functionalities.

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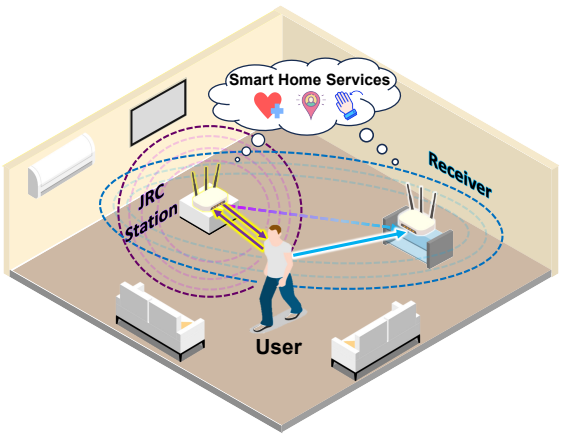


Fig. 1: The primary sensing tool is a mmWave radar. As gathering OFDM signals, the receivers simultaneously analyze CSI to enhance the accuracy of the sensing process.

B. Challenges and Solutions

This paper introduces *mSAC*, an innovative localization system that enhances precision through the integration of mmWave MIMO-OFDM sensing and communication. The core functionality of the mSAC system is illustrated in Fig. 1, where the Joint Radar-Communication (JRC) station transmits OFDM signals used for both radar sensing of the environment and communication purposes. These OFDM signals, characterized by known preamble sequences at both the JRC station and communication receiver sides, serve dual roles. For simplicity in subsequent discussions, the OFDM signals transmitted by the JRC station and reflected back to it are termed *radar signals*, while the same signals received by the communication receiver are referred to as *communication signals*. The communication receiver side is also abbreviated as *receiver side* and JRC station side as *JRC side*.

On one hand, the receiver side can decode the preamble sequences to obtain channel state information (CSI) information, which facilitates communication. On the other hand, CSI reflects the extent to which signals are affected by users in the environment and has proven to be an effective sensing information. Therefore, we need to integrate the information from both the JRC station and the receiver sides to enhance the accuracy of sensing. The novelty of mSAC lies in its ability to fuse sensing information from both the Radar and Communication perspectives, simultaneously improving resolution, range, and energy efficiency. This sets mSAC apart from existing OFDM mmWave ISAC systems, which focus primarily on either radar or communication, leaving their joint potential underutilized.

Challenge 1: How to build a model and extract effective features of OFDM signals? Adapting OFDM for sensing presents the challenge of accurately extracting effective features from communication signals. OFDM signals are suitable for data transmission, not sensing, which complicates their use for precise measurements. Contrasting with FMCW signals, which deliver high-resolution range and velocity data, OFDM necessitates innovative strategies to unlock their potential for accurate sensing. Through analysis of phase offsets, we have identified that time of flight (ToF) and angle of arrival (AoA) serve as effective features for the Radar sensing, while time difference of flight (TDoF) proves advantageous for the receiver side. Our comprehensive model integrates these features, leveraging ToF for precise distance measurements and AoA to pinpoint signal direction for the Radar side. Furthermore, TDoF enhances detection and localization accuracy in the receiver side, empowering robust sensing capabilities with OFDM signals.

Challenge 2: How to integrate Radar and Communication information? Integrating Radar and Communication analysis into a cohesive sensing framework is the other key challenge. Radar signal processing, leveraging ToF and AoA, yields high-resolution range-angle relationships but encounters limitations in terms of range. Conversely, the receiver side analysis, employing TDoF, offers broader coverage yet compromises on resolution. To strike a balance between these two aspects, we propose a solution where the sensing area is represented as a combination of a circle and an ellipse. The circle effectively captures high-resolution Radar features, offering precision in close proximity. In contrast, the ellipse encompasses broader the communication features, providing extended coverage. The simplest method of expanding the sensing range by adding an additional pair of transceivers results in overlapping circular zones that do not efficiently utilize radar information and prove to be not cost-effective. Our method seamlessly fuses Radar and Communication sensing features, with a focus on analyzing overlapping areas to enhance accuracy and reliability. This fusion approach improves sensing performance, demonstrating the capability of mSAC to serve as a comprehensive ISAC solution by fully leveraging spatial diversity.

C. Contributions

Our contributions can be summarized as follows:

- We develop a dual-end sensing fusion strategy that combines Radar and Communication signals into a unified model, achieving higher resolution for sensing and an extended range for coverage. We propose a theoretical device-to-device model that unifies sensing and communication capabilities. To validate our approach, we designed and conducted real-world experiments on a JRC platform, collected data, and tested our fusion strategy.
- Our experiments show a 65.93% increase in sensing accuracy and a 43.19% increase in range, validating the performance of the proposed solutions.

The remainder of this paper is structured as follows: Section II reviews related work in the field of tracking systems. Section IV provides the feature extraction and collaboration. Section V realizes the spectrum analysis and fusion. Section III introduces our proposed model. Section VI details our experimental setup and results. Finally, Section VII concludes the work of this paper.

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II. Related Works

In this section, we present an overview of the three primary categories of the state-of-art that are relevant to our research: FMCW-based localization, OFDM-based Wi-Fi localization, and integrated sensing and communication. FMCW is widely recognized for its precise range and velocity measurements by analyzing the frequency shift between transmitted and received signals. OFDM-based localization exploits existing Wi-Fi infrastructure to perform environmental sensing, utilizing the fine-grained channel state information to detect and track movements. ISAC synergistically combines communication and radar sensing capabilities, aiming to optimize both functionalities while sharing spectral resources.

A. FMCW-based mmWave Localization

Fang et al. [14] present a pedestrian trajectory tracking method using FMCW-MIMO radar, improving detection and tracking performance under low-observable conditions. Zhang et al. [15] demonstrate a pedestrian trajectory tracking system for autonomous urban driving, leveraging FMCW radar to achieve superior performance under low SNR conditions. Kumuda et al. [16] evaluate FMCW radar for multi-target detection and tracking, mitigating spot jammer attacks and improving tracking accuracy using CFAR detectors and clustering algorithms. Kang et al. [17] present a stepped-frequency hybrid modulation radar for target tracking, combining the benefits of Frequency-shift keying and chirp modes to improve range and displacement measurement accuracy.

FMCW radar technology, predominantly used for tracking and detecting targets in challenging environments, focuses on sensing and encounters communication inefficiencies in mmWave systems. These inefficiencies arise from limitations such as additional resource allocation, which requires additional channel estimation and synchronization leading to reduced efficiency [18], or the use of FMCW chirp as information carriers that results in mutual interference between communication and radar signals, increased bit error rates [19], and communication restrictions to low data rates [20]. In contrast, our approach employs OFDM mmWave radar and communication, significantly enhancing localization accuracy and ensuring robust communication by exploiting the spatial diversity between radar and communication modalities.

B. OFDM-based Wi-Fi Localization

IndoTrack [21] demonstrates that Wi-Fi passive tracking relies on detecting the Doppler frequency shift created by user movement through the Fresnel zone. Widar [22], [23] establishes the theoretical groundwork for device-free tracking by showing how to derive user velocities from CSI. HyperTracking [24] introduces a new concept for Wi-Fi passive tracking in non-line-of-sight scenarios, emphasizing that user movement through the hyperbolic zone is crucial, and develops a corresponding fundamental model.

However, due to the significantly higher frequencies and wider bandwidths of mmWave signals, their propagation characteristics differ substantially from those of lower-frequency Wi-Fi signals. These differences necessitate precise beamforming and dynamic tracking, which are not integral to traditional Wi-Fi-based localization techniques. In contrast, our approach utilizes both OFDM mmWave radar and communication CSI to simultaneously facilitate localization and communication, offering a distinct methodology from traditional OFDM-based Wi-Fi localization techniques.

C. Integrated Sensing and Communication

Sturm et al. [25] introduced and validated an OFDM system design capable of conducting both radar and communication tasks in the 24 GHz band, demonstrating a notably high dynamic range for radar imaging. In [26], they explored range and angle estimation methods using MIMO-OFDM waveform design to match the array aperture and full signal bandwidth. Saddik et al. [19] design and implement an ultra-wideband JRC system that simultaneously transmits communication and radar signals by leveraging the quasi-orthogonality of up-chirp and down-chirp of linear FMCW. In [27], a MIMO Joint Radar Communication testbed operating in the 24 GHz was showcased, utilizing an OFDM waveform to simultaneously transmit data across all subcarriers. Additionally, the development of a real-time, full-duplex JRC testbed equipped with a fully-digital front-end and software-defined radios was highlighted, significantly enhancing the JRC's flexibility and capabilities. De et al. [28] explore how 6G technologies integrate these components to enhance opportunities while addressing new challenges. Long et al. [29] emphasize the importance of Line-of-Sight detection and accurate parameter estimation for effective localization. Yu et al. [30] discuss optimizing beamforming to improve location sensing for multiple users in intelligent reflecting surface-enabled systems. Zhang et al. [31] investigate the balance between communication throughput and sensing accuracy in cooperative ISAC systems. Lastly, Destino et al. [32] analyzes the critical trade-offs between high data rate communication and precise positioning in mm-wave communications, vital for ISAC applications.

Despite existing advancements, current ISAC systems do not fully utilize the spatial diversity available in sensing and communication. In contrast, our work addresses this gap by introducing a dual-end fusion approach in mSAC, which combines radar and communication features to leverage spatial diversity, to enhance both the accuracy and range of localization.

III. Integrated Sensing Modeling

In this section, we present the mathematical model used to describe the geometry and interactions of the elements in our system. The equations define the shapes and their intersections, leading to the calculation of areas and enhancements in the sensing region.

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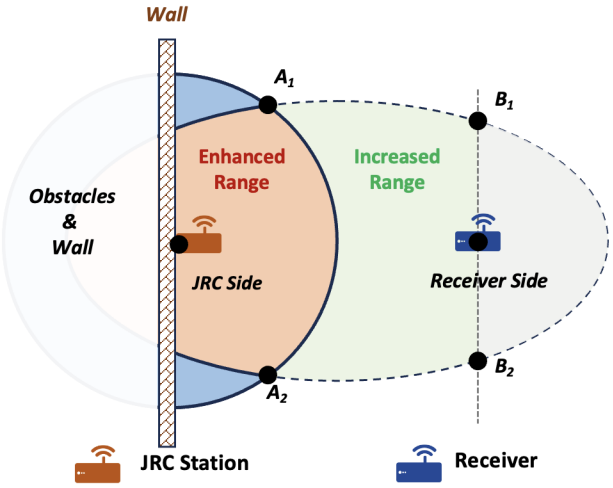


Fig. 2: Illustration of the geometric model, showing the definitions of key variables such as Pc_1, Pc_2, and their relationship with the circle and ellipse.

For a given object, assuming the effective sensing radius is r, the sensing range of the JRC station can be modeled as a circle with a radius of r. For the receiver side, the sensing range satisfies the following equation:

$$|Pc_1| + |Pc_2| - |c_1c_2| < 2r - |c_1c_2|, \tag{1}$$

where P denotes the user's location, characterizes an ellipse, confirming that the combined distances from points c_1 and c_2 to P minus their separation is less than a fixed value, thus conforming to the properties of an ellipse. Here, Pc_1 and Pc_2 represent the distances from P to the foci of the ellipse, c_1 and c_2, respectively, as shown in Fig. 2. These relationships clarify the geometric structure of the sensing region. This formulation allows the receiver's sensing area to be modeled as an ellipse, ensuring that the reflected path does not exceed d.

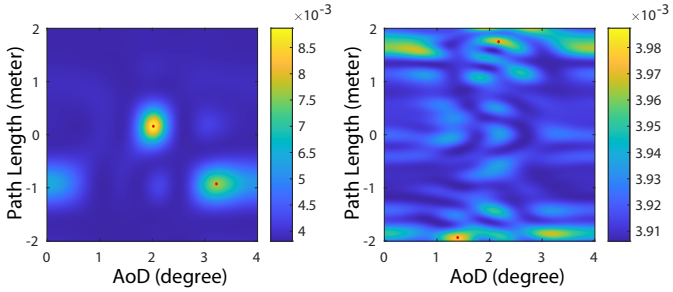
When we determine the TDoF, the target is located at the ellipse. In a real-world scenario, JRC station equipment typically covers only the direction it is facing. Therefore, we adjust the sensing area as shown in Fig. 2. When we calculate the ToF, the target is located at the circle.

Mathematically, we can utilize the following model to represent the circle and the ellipse:

$$\begin{cases} x^2 + y^2 = r^2, (x > 0), \\ \dfrac{(x - \frac{d}{2})^2}{r^2} + \dfrac{y^2}{r^2 - (\frac{d}{2})^2} = 1, (0 < x < d) \end{cases} \tag{2}$$

where x and y are the coordinates in a Cartesian plane, r is the effective sensing radius, and d is the distance between the JRC station side and the receiver side. According to the definition of an ellipse, we know that $a = r$, and $c = d/2$. Consequently, we have $b^2 = a^2 - c^2 = r^2 - (d/2)^2$.

The colored regions in Fig. 2 can be divided into three parts: i) the green area denotes the increasing sensing area; ii) the red area denotes the overlapping sensing area; iii) the blue area denotes the original sensing area.



(a) With PDD removal. (b) Without PDD removal.

Fig. 3: Simulation of PDD removal impact.

In this model, we assume that the AoD at the transmitter is equivalent to the AoA at the receiver side under line-of-sight (LoS) conditions. This assumption simplifies the analysis by leveraging the symmetry of signal paths in LoS scenarios.

We can observe that there is an opportunity to utilize the new receiver to increase the sensing distance and enhance the accuracy of perception within overlapping areas.

To calculate the intersection points of the circle and the ellipse, we derive and solve the following quadratic equation in x:

$$dx^2 - (d^2 - 4r^2)x - 2dr^2 + \frac{d^3}{4} = 0, \tag{3}$$

Solving this equation, we obtain two intersection points (A_1 and A_2) given by the coordinates:

$$\left(\frac{d}{2}, -\sqrt{r^2 - \frac{d^2}{4}}\right) \quad \text{and} \quad \left(\frac{d}{2}, \sqrt{r^2 - \frac{d^2}{4}}\right). \tag{4}$$

Meanwhile, to determine the blind area of the receiver, we can obtain two intersection points (B_1 and B_2) denoted by:

$$\left(d, r - \frac{d^2}{4r}\right) \quad \text{and} \quad \left(d, -r + \frac{d^2}{4r}\right). \tag{5}$$

Based on the above intersection points, we can calculate the increasing sensing area denoted by S_{new}, based on the intersection of the ellipse and the circle. Specifically, this area is determined using the integral:

$$\begin{aligned} S_{new} = &\int_{-\sqrt{r^2 - \frac{d^2}{4}}}^{\sqrt{r^2 - \frac{d^2}{4}}} \left(r\sqrt{1 - \frac{y^2}{r^2 - (\frac{d}{2})^2}} + \frac{d}{2} - \sqrt{r^2 - y^2}\right) dy \\ &- \int_{-r + \frac{d^2}{4r}}^{r - \frac{d^2}{4r}} \left(r\sqrt{1 - \frac{y^2}{r^2 - (\frac{d}{2})^2}} - \frac{d}{2}\right) dy. \end{aligned} \tag{6}$$

We calculate the area of the ellipse (S_{elli}) as:

$$\begin{aligned} S_{elli} = &\int_0^{r + d/2} 2\sqrt{\left(1 - \frac{(x - \frac{d}{2})^2}{r^2}\right)\left(r^2 - \frac{d^2}{4}\right)} dx \\ &- \int_{-r + \frac{d^2}{4r}}^{r - \frac{d^2}{4r}} \left(r\sqrt{1 - \frac{y^2}{r^2 - (\frac{d}{2})^2}} - \frac{d}{2}\right) dy, \end{aligned} \tag{7}$$

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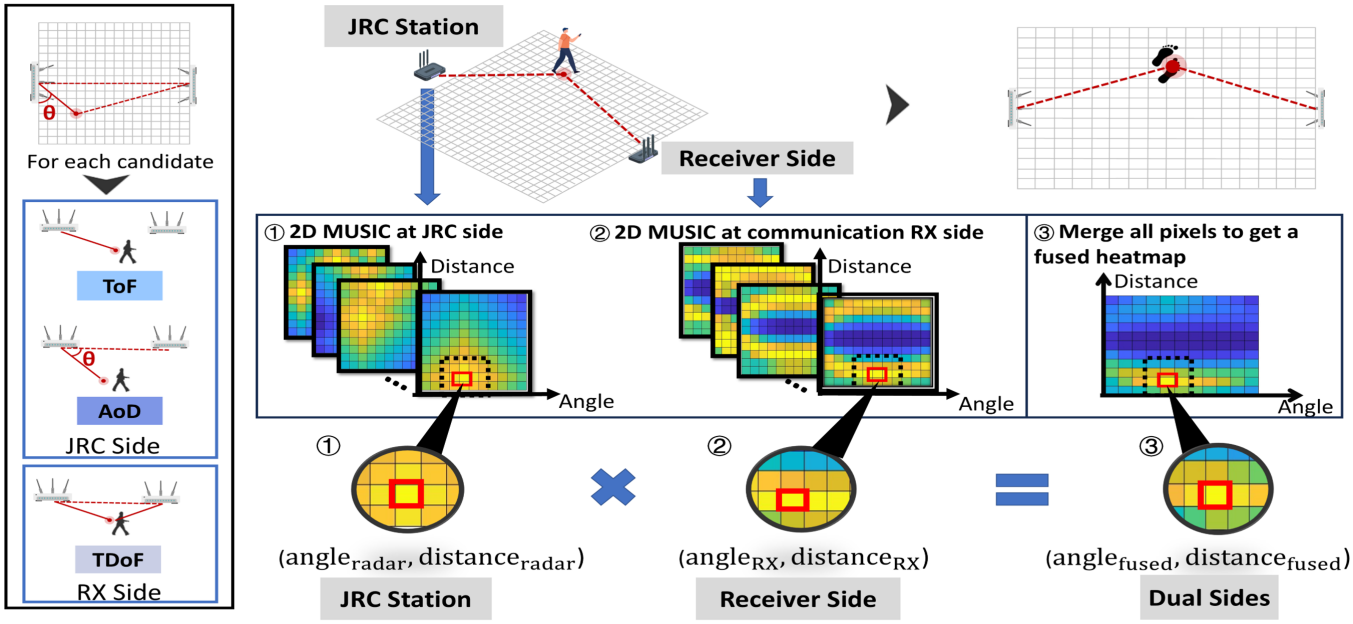


Fig. 4: Illustration of Heatmap Analysis and Fusion: The 2D-MUSIC method is applied independently on both the JRC and Receiver sides. Subsequently, the results are merged cell by cell to produce a comprehensive fused heatmap.

where x represents the horizontal coordinate in the plane, and the integral calculates the total area by integrating over the horizontal span of the ellipse.

The enhancement in the sensing area, S_{enh}, is defined as the difference between the area of the ellipse and the new area created by the intersection:

$$S_{enh} = S_{elli} - S_{new}. \tag{8}$$

To optimize the receiver setup, minimizing additional distances ensures efficient coverage while reducing redundant overlaps. This approach enhances the overall sensing efficiency and accuracy. This model provides a comprehensive framework for analyzing the geometric relationships in the sensing region, improving the sensing performance.

IV. Feature Extraction and Collaboration

This section describes the methods for extracting AoD and ToF features from both the radar and receiver sides, addressing key challenges such as phase offsets, signal estimation, and error mitigation.

A. Phase Offsets in Feature Extraction

Phase offsets are a key issue affecting the accuracy of parameter estimation, such as AoA, AoD, and ToF. They are typically caused by system design flaws or hardware limitations. Two common types of phase offsets are Initial Phase Offset (IPO) and Packet Detection Delay (PDD). IPO arises from phase mismatches between antennas, introducing errors in angle estimation and reducing spatial accuracy [33], [34]. PDD results from hardware timing errors, disrupting synchronization and degrading ToF precision, especially in asynchronous systems. These offsets complicate signal processing, making it crucial to develop effective solutions for reliable feature extraction in various applications.

B. 2-Dimensional MUSIC

The 2-dimensional Multiple Signal Classification (MUSIC) algorithm is a high-resolution technique for estimating AoA and ToF. It is foundational for feature extraction, enabling precise localization and signal parameter estimation.

The algorithm constructs a spatial spectrum from the covariance matrix of received signals, identifying peaks for AoD and ToF. These peaks correspond to the estimated parameters of interest, with higher spectrum values indicating potential signal directions or delays.

For AoD estimation, the algorithm projects steering vectors onto the noise subspace, resulting in a power spectrum that identifies the signal directions:

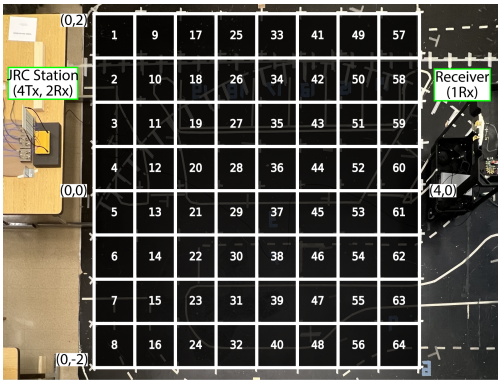
$$P_{music}(\theta) = \frac{1}{a^H(\theta) E_N E_N^H a(\theta)}, \tag{9}$$

where P_{music} represents the MUSIC power spectrum, $a(\theta)$ is the steering vector for the angle of departure, and E_N is the noise subspace eigenvector matrix derived from the signal covariance matrix. Peaks in $P_{music}(\theta)$ indicate the AoD estimates.

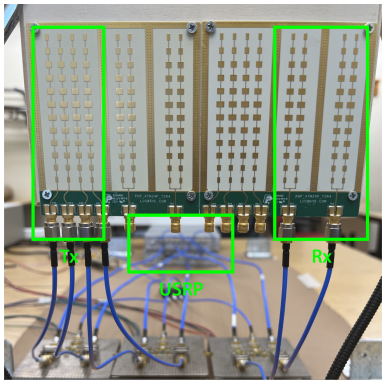
For ToF estimation, a joint 2D spectrum is created by combining distance and angle steering vectors using the Kronecker product. This approach accounts for both spatial and temporal signal characteristics:

$$P_{\text{music_2D}}(r, \theta) = \frac{1}{a(r,\theta)^H E_N E_N^H a(r,\theta)}, \tag{10}$$

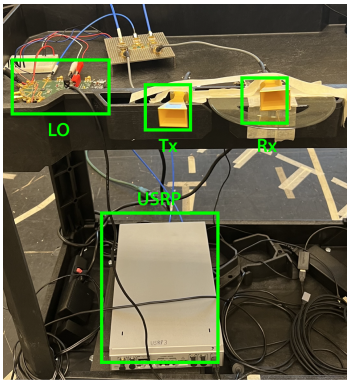
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(a) A setup of devices and sensing areas.



(b) Device layout at the JRC Station.



(c) Device layout at the Receiver Side.

Fig. 5: Experimental setup.

where $a(r,\theta)$ is the joint steering vector that combines distance and angle information:

$$a(r,\theta) = a_{dist}(r) \otimes a_{angle}(\theta). \tag{11}$$

The distance steering vector $a_{dist}(r)$ models the time delays for different propagation distances:

$$a_{dist}(r)[m_{sub}] = \exp\left(-j2\pi m_{sub}\Delta f \frac{2r}{c}\right), \tag{12}$$

where m_{sub} is the subcarrier index, c is the speed of light, $\Delta f = B/N_{subcarrier}$ is the subcarrier frequency spacing, and r is the propagation distance.

Similarly, the angle steering vector $a_{angle}(\theta)$ models phase shifts across antennas based on the AoD:

$$a_{angle}(\theta)[m_{ant}] = \exp\left(j2\pi f_c m_{ant} \frac{d\sin(\theta)}{c}\right), \tag{13}$$

where m_{ant} is the antenna index, f_c is the carrier frequency, d is the antenna spacing, and θ is the angle of departure.

While MUSIC is effective for distinct multipath signals with fewer sources than RF chains, its performance degrades when handling correlated signals or when the number of sources exceeds the available RF chains. This limitation underscores the need for advanced methods in more complex environments.

C. JRC Side Analysis

The JRC side uses 2D MUSIC to extract AoA and ToF features from radar signals, addressing challenges posed by IPO and PDD. Null data packets with preamble symbols are transmitted orthogonally across antennas to create an omnidirectional radiation pattern suitable for MIMO radar processing.

1) AoA Analysis: The MUSIC algorithm is widely used for AoA estimation. It calculates a spatial spectrum by projecting steering vectors onto the noise subspace:

$$P_{music}(\theta) = \frac{1}{a^H(\theta) E_N E_N^H a(\theta)}, \tag{14}$$

where P_{music} is the MUSIC power spectrum for the signal arriving from direction θ, $a(\theta)$ is the steering vector representing the phase shifts across the antenna array, and E_N is the noise subspace eigenvector matrix. This method effectively resolves AoA by identifying the direction that maximizes the power spectrum.

2) ToF Analysis: For ToF estimation, the MUSIC algorithm extends to a 2D temporal spectrum by applying the Kronecker product of distance and angle steering vectors:

$$P_{\text{music_2D}}(r,\theta) = \frac{1}{a(r,\theta)^H E_N E_N^H a(r,\theta)}, \tag{15}$$

where $a(r,\theta)$ is the joint 2D steering vector representing the propagation distance r and angle θ.

This approach enables precise feature extraction for both AoA and ToF, effectively mitigating inaccuracies caused by Initial Phase Offset (IPO) and Packet Detection Delay (PDD).

D. Receiver Side Analysis

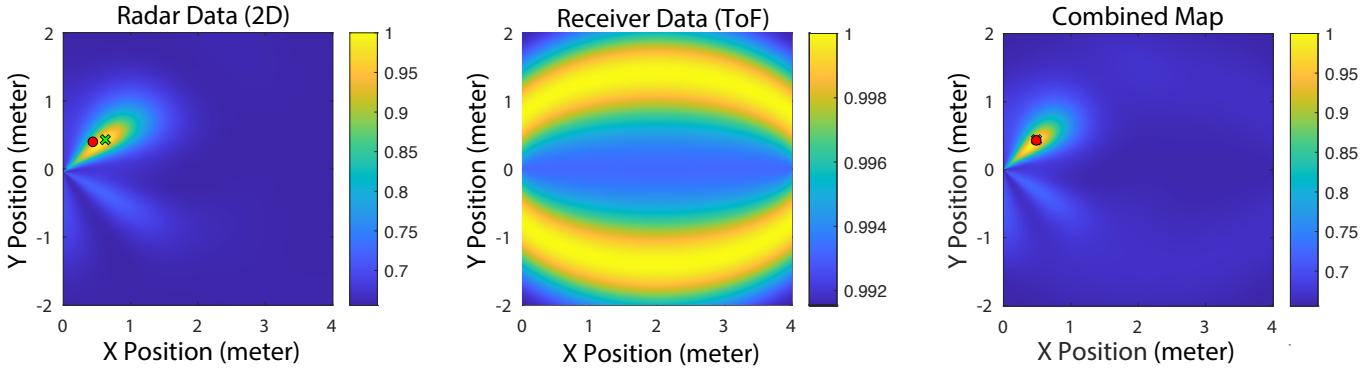
1) AoD Estimation: The receiver side uses a simplified setup with a single antenna to estimate the transmitter's AoD instead of AoA. This approach reduces hardware complexity and leverages the stronger capabilities of JRC stations. The AoD is estimated by analyzing the phase shifts at the transmitter and correlating them with the spatial spectrum, enabling precise departure angle determination with minimal hardware requirements.

PDD introduces phase shifts that reduce localization accuracy. To mitigate this, we employ a ToF sanitization algorithm that removes linear phase offsets caused by Symbol Timing Offset (STO). The algorithm estimates and corrects the phase offsets as follows:

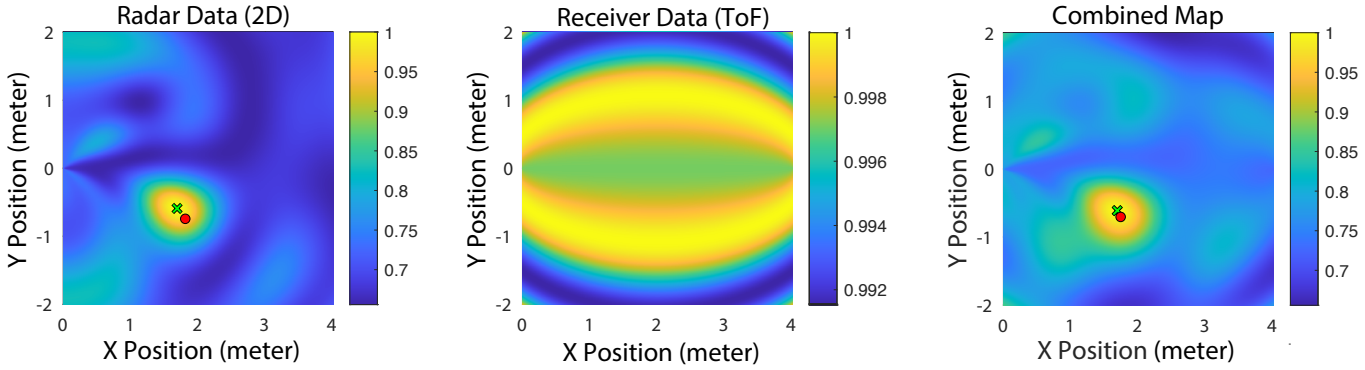
$$\hat{\tau} = \arg\min_{\tau} \sum (\phi + 2\pi\Delta f_{\text{list}}\tau)^2, \tag{16}$$

where $\hat{\tau}$ is the estimated time delay, ϕ represents the observed phase values, and Δf_{list} is the subcarrier frequency interval. This equation minimizes the squared error between

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(a) Example 1: An individual at Location 4: (0.25, 0.25).



(b) Example 2: An individual at Location 30: (1.75, -0.75).

Fig. 6: Examples of *mSAC* localization results from the JRC station's radar side, Receiver side, and their combination.

the observed phases and the modeled linear phase offset, identifying the optimal τ.

$$\hat{\phi} = \phi + 2\pi\Delta f_{\text{list}}\hat{\tau}, \tag{17}$$

where $\hat{\phi}$ is the corrected phase, ϕ is the unwrapped phase, and the term $2\pi\Delta f_{\text{list}}\hat{\tau}$ accounts for the estimated linear phase offset. This correction ensures that the phase data aligns accurately across subcarriers.

As illustrated in Fig. 3, simulation results demonstrate that removing PDD significantly enhances localization accuracy by distinguishing the TDoF of reflected and direct paths. Without PDD removal, localization errors increase substantially, impairing the system's ability to differentiate signal paths effectively.

V. Heatmap Analysis and Fusion

We now have two heatmaps, one from the JRC side and one from the receiver side. The JRC side represents the correspondence of AoA and ToF, while the receiver side represents the correspondence of AoD and DToF. As highlighted in Fig. 4, the 2D-MUSIC method is independently applied to both sides to generate initial heatmaps. These are then merged cell by cell to create a fused heatmap, which enables a more precise localization of the target.

It is worth noting that the AoA from the JRC side and the AoD from the receiver side refer to the same physical angle, ensuring consistency between the two heatmaps. This consistency is also depicted in Fig. 6, where the relationship between AoA, AoD, and their respective ToF and DToF measurements is illustrated. The star in Fig. 6 indicates the actual target location, while the red point shows the predicted target position based on initial estimations.

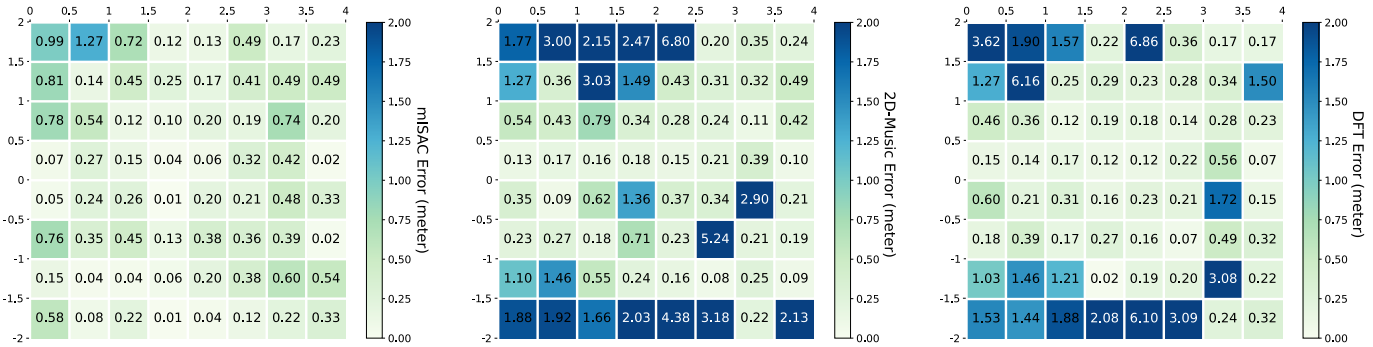
In Fig. 4, we focus on the process of heatmap fusion using the 2D-MUSIC method, which complements the perspective. While Fig. 6 explains the theoretical alignment between AoA and AoD measurements, Fig. 4 demonstrates how the practical fusion of radar and communication signals results in a unified and accurate heatmap for target localization.

A. Combining Radar and Communication Localization Results

To achieve precise localization, we integrate the positions estimated by both the communication and radar systems. Let $\vec{p}_{\text{re}}(t)$ and $\vec{p}_{\text{ra}}(t)$ denote the positions estimated by the receiver side and radar systems at time t, respectively. As the heatmaps generated in Fig. 4 provide an intermediate step, these positions are obtained through cell-by-cell fusion, which is then mapped into real-world coordinates.

Since the coordinate systems and sampling intervals differ between the communication and radar systems, we apply coordinate transformations T_{re} and T_{ra} to map $\vec{p}_{\text{re}}(t)$ and $\vec{p}_{\text{ra}}(t)$ into a common coordinate system. This ensures consistency and alignment:

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(a) mSAC. (b) 2D-Music. (c) DFT.

Fig. 7: Overall localization performance of *mSAC*, 2D-Music, and DFT.

$$\vec{p}'_{\text{re}}(t) = T_{\text{re}}\vec{p}_{\text{re}}(t), \quad \vec{p}'_{\text{ra}}(t) = T_{\text{ra}}\vec{p}_{\text{ra}}(t). \tag{18}$$

The transformed positions are combined into a single estimate $\vec{p}(t)$ using a weighted approach that reflects the reliability of each system. As shown in Fig. 4, the weights are determined by visualizing the relative confidence of each cell in the fused heatmaps.

This methodology builds directly on Fig.4, which highlights the contributions of individual systems and their integration during fusion. In contrast, Fig.6 focuses on the theoretical framework, illustrating angle and time correspondence without detailing the cell-level heatmap fusion process.

VI. Evaluation

A. Experiment Setup

1) Testbed Implementation: We utilize a mmWave MIMO OFDM joint radar communication testbed platform, a twin architecture to the one in [35], to establish a radar-communication station with a configuration of 4TX-2RX and a communication receiver equipped with a single horn antenna. The baseband signal processing is provided by the USRP N32x series from National Instruments, and these USRP units are synchronized using an external clock distributor provided by National Instruments to facilitate MIMO operations. For frequency conversion, we employed a local oscillator (LO) from Analog Devices and a custom up/down converter circuit from X-Microwave, responsible for upconverting and downconverting signals between the baseband and 24 GHz. The baseband signals were transmitted to and from the PC using an Intel X520 network card connected to the USRP.

The 4TX-2RX channel was facilitated by the MIMO antenna array from Luswave Technology, with each transmitter array separated by 6.35 mm and each receiver array by 25.4 mm, and the maximum achievable transmit power is 21dBm. We implemented a carrier frequency of $f_c = 24$ GHz and a bandwidth of B =125 MHz, incorporating N_{sc} =64 OFDM subcarriers. Additionally, Quadrature Phase Shift Keying (QPSK) was employed as the modulation scheme. The choice of 64 subcarriers in the experiment setup reflects a balance between resolution and computational efficiency. Using 64 subcarriers provides sufficient frequency granularity to capture multipath effects without overwhelming system resources, making it suitable for OFDM-based implementations.

2) Dataset: In our study, we organize the data collection within a 4m x 4m sensing area, segmented into 64 blocks, each 0.5m x 0.5m. For each block, we collected three data sets, each consisting of 300 packets containing measurements and signals relevant to our study. Specifically, we captured data while an individual were positioned at the center of each block. It is notable to emphasize that all results presented in this section are based on real data.

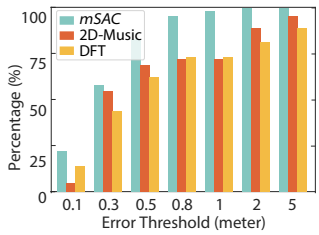
Furthermore, all comparison algorithms employed in this analysis utilize the same dataset, exclusively drawing from data collected at the JRC station. This consistency ensures a fair comparison across different methodologies, highlighting the effectiveness of our approach. This arrangement ensures a detailed and consistent dataset for analysis.

B. Experiments and Results

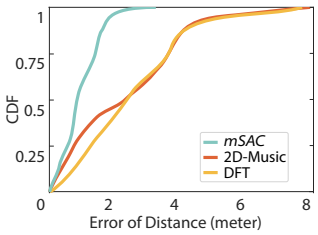
1) Overall Performance: To illustrate the enhancements *mSAC* brings to localization accuracy, we present two specific examples in Fig. 6. On the JRC side, a radar heatmap is generated that illustrates the estimated locations of targets, which may not align precisely with the ground truth due to errors and imperfections in channel estimation. Simultaneously, the Receiver side produces a separate heatmap, primarily focused on Time of Flight (ToF) analysis, which results in an elliptical representation that further highlights the influence of ToF variations on localization accuracy.

By applying the *mSAC* fusion model, these individual heatmaps are integrated, leveraging the spatial diversity inherent in combining radar and communication data. This integration helps to mitigate the inaccuracies caused by channel estimation errors and other imperfections. The fusion process culminates in more refined predictions, where stars in the figures represent the ground truth locations of the targets, and dots indicate the position predictions. This enhanced method clearly demonstrates the substantial accuracy improvements provided by the fusion model, showcasing

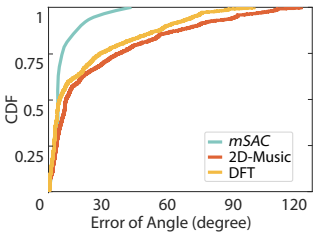
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(a) Localization accuracy.



(b) Accuracy of distance.



(c) Accuracy of angle.

Fig. 8: Cumulative error distribution for *mSAC*, 2D-Music, and DFT.

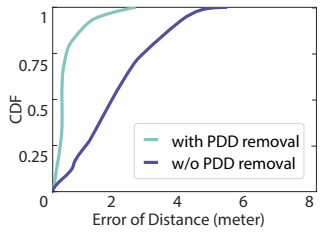
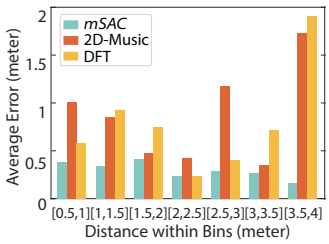
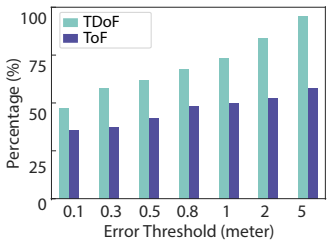


Fig. 9: Benifits of PDD removal.



(a) Impact of Distance.



(b) Comparison of TDoF and ToF.

Fig. 10: Parameter Study.

how effectively *mSAC* utilizes both radar and communication systems to achieve superior localization performance.

2) Localization Accuracy: Figure 7(a) presents the overall performance of *mSAC*, achieving an average localization error of 0.31 meters. This improvement is attributed to both the increased number of antennas and the synergy of our theoretical model, which expands the sensing range and coverage area. We benchmark *mSAC* against state-of-the-art methods like 2D-MUSIC [35] and DFT [36]. Fig. 7(b) and 7(c) depict their system heatmaps. Notably, errors increase with both the angle and distance from the central router position, highlighting the efficacy of *mSAC*'s advanced sensing capabilities. As shown in Fig. 8, *mSAC* outperforms these methods, exhibiting 65.93% and 69.00% improvements in localization accuracy over 2D-MUSIC and DFT, respectively, which correspond to errors of 0.91 meters and 1.00 meters. These enhancements are largely due to the extended sensing range beyond the limitations of the traditional circle model employed by both comparative systems.

3) Benefits of PDD removal: The PDD significantly impacts system performance, as illustrated in Fig. 3. Without PDD removal, the localization phase cannot distinguish phase offsets, leading to substantial performance degradation. Experimental results indicate a higher error rate in localization accuracy, as the system struggles to differentiate between actual signal paths and delayed ones. This inability to accurately detect and compensate for path delays results in incorrect measurements and reduced overall system precision.

C. Parameter Study

1) Distance from the Origin: To evaluate whether *mSAC* is sensitive to distance—specifically, if its performance deteriorates as the distance increases—we divided the sensing area into several bins based on their distance from the origin. As shown in Fig. 10(a), we also compared *mSAC* with 2D-MUSIC and DFT to assess their sensitivity to distance. The results reveal that *mSAC* maintains relatively uniform performance across distances, whereas 2D-MUSIC and DFT exhibit significant fluctuations as the distance increases. Moreover, *mSAC* significantly enhances the effective sensing area compared to the alternative methods. For instance, in the range of 3.5 to 4 meters, both 2D-MUSIC and DFT exceeded an error margin of 1.5 meters, whereas *mSAC* achieved a much lower error of just 0.23 meters.

2) ToF vs. TDoF: This section investigates the effectiveness of using TDoF instead of ToF on the receiver side. As illustrated in Fig. 10(b), the performance of both methods is compared under various error thresholds. The results clearly demonstrate that TDoF consistently outperforms ToF. This advantage arises from the inability of ToF to completely separate reflected signals from the mixed signals collected, thereby limiting its ability to accurately determine the ToF of reflection paths. In contrast, TDoF measurements remain precise even in the presence of inseparable reflection paths. This robustness makes TDoF particularly suitable for environments with complex signal reflections. Consequently, this experiment confirms that TDoF provides more reliable and accurate information about the target.

VII. Conclusion

In this paper, we introduced *mSAC*, a system that combines mmWave-based joint sensing technologies into a unified framework using OFDM. This integration not only addresses the limitations of traditional FMCW and OFDM systems but also improves accuracy and range. We explored the technical challenges of integrating sensing and communication functionalities, proposing methods that enhance the system's efficiency. Through advanced fusion techniques and optimization of the OFDM framework, our evaluations show that *mSAC* achieves substantial improvements in sensing accuracy by 65.93% and an increase in the sensing range by 43.19%. These enhancements make *mSAC* particularly effective compared to existing methods. In conclusion, *mSAC* effectively advances mmWave-based technologies, offering a robust, accurate, and versatile platform for future developments in integrated sensing and communication systems.

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